

bps, a year forward was around 100 bps, and two years forward was around 80 bps.³⁴

Consider now an investor who is presented with the vol and swap curves in this figure and has the view that neither curve is going to change over the horizon of a year. Suppose the investor purchases a linear curve cap on 2s10s that expires in two years and is struck at-the-money forward. This cap pays out 1 bp of the notional for each bp that 2s10s is steeper than the strike at expiry. That is, the payout on this option in two years is

$$\max[(10\text{-year CMS} - 2\text{-year CMS}) - \text{Strike}, 0] \times \text{Notional}. \quad (9.9)$$

Assume this option can be purchased by the investor for 33 bps of the notional. Suppose now the investor holds on to the option for one year (at which time the option has one year remaining in its life). Consider the change in the value of the option over this time frame under the assumption that the swap curve and vol curve are both unchanged in one year. The change in the value of the option is displayed in Table 9.6.

What's going on here? The investor buys the option for 33 bps, and a year later it is worth 37.5 bps. We usually think of the value of options going down as time to expiry draws near and the market doesn't move due to the time decay of the option, but the value of this option is actually going up. "These go to eleven."³⁵ The time decay of the option is certainly a factor

³⁴It is worth repeating that it would be most appropriate to look at convexity-adjusted forward rates in the analysis of curve options. We will continue to ignore such adjustments for our purposes here.

³⁵In the movie *This Is Spinal Tap*, musician Nigel Tufnel (played by Christopher Guest) is being interviewed by Marty DiBergi (played by Rob Reiner). Nigel is showing his amplifier to Marty and points out that the volume knobs all go to a highest setting of 11 (as compared to most amplifiers, whose volume knobs only go up to 10). Nigel then claims that this numerical schema means that his amp is louder ("It's one louder"). When Marty suggests that the volume knobs could be renumbered so that 10 is loudest, Nigel, who is clearly confused, replies after a bit, "These go to eleven." The excerpt from the movie is as follows:

Nigel: ... the numbers all go to eleven. Look ... right across the board.
 Marty: Ahh ... oh, I see. ...
 Nigel: Eleven ... eleven ... eleven. ...
 Marty: ... and most of these amps go up to ten. ...
 Nigel: Exactly.
 Marty: Does that mean it's ... louder? Is it any louder?
 Nigel: Well, it's one louder, isn't it? It's not ten. You see, most ... most blokes, you know, will be playing at ten. You're on ten here ... all the way up ... all the way up. ...
 Marty: Yeah...
 Nigel: ... all the way up. You're on ten on your guitar ... where can you go from there? Where?
 Marty: I don't know. ...

Initial cost of curve cap	33.0 bps
After one year	
Time value (theta)	-7.6 bps
Curve change (delta)	+9.6 bps
Vol change (vega)	+2.5 bps
Value of curve cap in one year, assuming rates/vol unchanged	37.5 bps

Table 9.6: Carry and Roll-Down Analysis on 2s10s Curve Cap

here, and according to Table 9.6 the value of that time decay is 7.6 bps after one year, assuming the vol curve and swap curve do not change. That is, assuming the vol and swap curves do not change over the course of the next year, the value of the option would decline 7.6 bps due to theta decay.

But there are two other factors at work here that counteract the theta bleed. First, recall from Figure 9.16 that 2s10s, two years forward is about 80 bps, but one year forward it's about 100 bps. That is, if the curve does not change at all over the course of a year, the underlying 2s10s curve will steepen about 20 bps. That is a good thing for the holder of a curve cap. Given the curve cap in question, in one year the option will have one year remaining in its life, and according to Table 9.6, the option will increase in value by 9.6 bps due to this approximately 20 bps steepening of the curve.³⁶ Second, when first purchased the option has two years until expiry, and thus after one year it has only one year until expiration. But according to Figure 9.16 the implied vol used to price one-year curve options is significantly higher than that used to price two-year curve options. According to Table 9.6, the change in the value of the option when it has one year remaining due to using the higher implied vol is +2.5 bps. As shown in Table 9.6, the net effect of these forces is that despite the loss of one year in the life of the option, the value of the option actually increases 4.5 bps.

Nigel: Nowhere. Exactly. What we do is if we need that extra ... push over the cliff ... you know what we do?

Marty: Put it up to eleven.

Nigel: Eleven. Exactly. One louder.

Marty: Why don't you just make ten louder and make ten be the top ... number ... and make that a little louder?

Nigel: ... these go to eleven.

From *This Is Spinal Tap*, released on March 2, 1984. Used by permission of Studio Canal.

³⁶Remember this is an option, and its delta in general will be less than one. Therefore, although the curve steepens 20 bps, this does not mean that the value of the option will increase 20 bps. It will increase by a lesser amount.

According to this “carry and roll-down” analysis, buying this option and holding it for a year is a pretty attractive proposition if neither the vol curve nor the swap curve changes shape. Of course, most investors don’t literally believe that these assumptions will hold true. Most will probably buy the curve option because they believe the curve will steepen. But if the investor is wrong, this analysis indicates the option should perform pretty well for the investor in case the market doesn’t change much over the course of the year.³⁷

9.3.7 Delta Hedging Curve Options

Recall that as more and more non-inversion notes were printed, dealers were getting longer and longer low-strike binary curve floors. As there was little demand from investors for these types of options, dealers most often sat on them. They had bought the options at relatively low (i.e., attractive) levels of implied volatility and would hedge out the risks associated with the positions as best as they could. The risk of prime importance was delta, which in the case of a curve option means the change in the value of the option with respect to a change in the steepness or flatness of the underlying curve.³⁸

Let’s consider a particular example. Define $S = 30\text{-year CMS} - 10\text{-year CMS}$. Consider a binary curve floor that pays off a fixed coupon 7% when S is less than 0 and 0% otherwise. When we think about delta hedging a curve option, it may be easier just to think of S as the sole underlying variable itself and not focus on the fact that it’s actually the difference between two random variables. The payoff at expiry of this option is shown in Figure 9.7. When S is positive, this option pays out zero, and when it’s negative, the option pays out a 7% coupon. This payoff profile looks very similar to the binary options we examine in Appendix A, except of course that the underlying random variable S can take on negative values in addition to positive ones. When S gets bigger, that means the curve is steeper, and when it’s smaller, that means the curve is flatter. And when S is negative, that means the curve is inverted.

In Appendix A, we show that the delta profile of binary options is unusual. In the case of a binary put on a stock, when the underlying is high and the option is out-of-the-money, as the underlying goes down and approaches

³⁷This type of analysis does not take into account a number of factors that could decrease the value of the options. For example, if curve option vol comes down and/or the (forward) swap curve flattens over the next year, the value of the curve option will decrease.

³⁸Although the most appropriate delta hedge might well consist of forward starting swaps, some dealers would instead use spot starting swaps for delta hedging curve options inasmuch as they have slightly better liquidity. For ease of exposition, we will consider the use of spot starting swaps when we speak of delta hedging curve options.

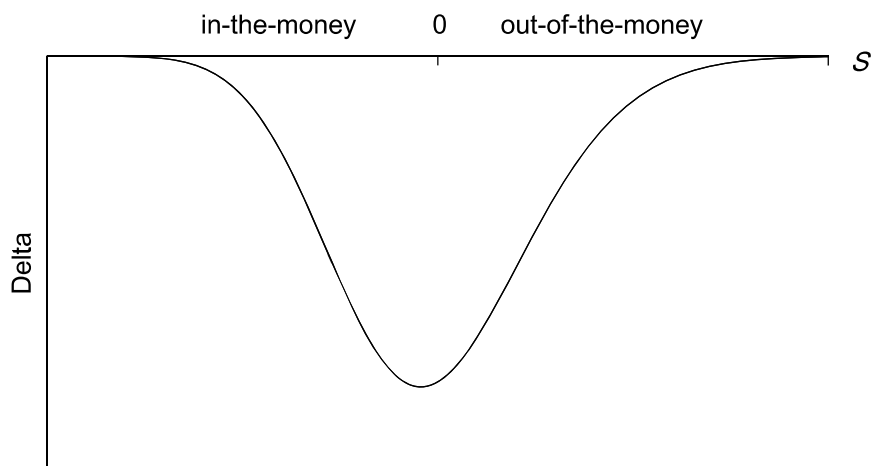
the strike, a trader who is long the option and delta hedging will be buying stock. And provided the stock doesn't cross below the strike and starts going back up, a delta hedger will sell some of the stock. So, in summary, provided the underlying remains above the strike, a delta hedger who is long the binary put will buy stock as the underlying goes down and will sell stock as the underlying goes up. But if the underlying should fall below the strike, then there is a dramatic change in the delta hedging activities of the trader. Once the underlying crosses below the strike, things work in reverse. With the underlying below the strike, a further decline in the underlying will cause a delta hedger to sell some of the underlying. And if the underlying should rise (but still remains below the strike), the delta hedger will buy stock. As we discussed in the Appendix A, the seemingly perverse change in the delta hedging activities are due to the fact that the binary put option has positive gamma when the option is out-of-the-money and negative gamma when the option is in-the-money.

The same logic applies to a binary curve floor. Suppose we are long an out-of-the-money binary curve floor and are delta hedging the position. Then our delta hedging activities will be as follows:

1. Provided that the curve doesn't ever flatten too much and the option remains out-of-the-money, then we will put on flatteners when the curve steepens and put on steepeners when the curve flattens.
2. If the curve should ever flatten enough such that the option is in-the-money, then we will put on flatteners when the curve flattens and put on steepeners when the curve steepens (provided, of course, that the curve doesn't steepen so much such that the option is again out-of-the-money, in which case we revert to the above case).

Assuming that the size of the option position is large enough, the delta hedging of the position can have significant implications on curve dynamics. Provided the curve is steep and the option is out-of-the-money, the delta hedging activities of someone who is long this option will actually tend to be a stabilizing force on the curve. That's because as the curve flattens, a delta hedger who comes in and puts on steepeners will cause the curve to steepen (on the margin) in the face of the flattening.³⁹ And as the curve steepens, delta hedgers will come in and put on flatteners, which on the margin will cause the curve to flatten. This was exactly the dynamic that contributed to

³⁹If an investor puts on a steepener in swaps, then he receives in the short-tenor swap and pays in the long-tenor swap. This will, on the margin, put downward pressure on the short-tenor swap rate and upward pressure on the long-tenor swap rate, which means that the delta hedger's activities will put pressure on the curve to steepen, counteracting the flattening that was prevailing.



Binary curve floor in-the-money:

1. Negative gamma profile
2. As curve flattens, delta hedger puts on flattener
3. As curve steepens, delta hedger puts on steepeners
4. Thus, delta hedging destabilizes curve

Binary curve floor out-of-the-money:

1. Positive gamma profile
2. As curve flattens, delta hedger puts on steepeners
3. As curve steepens, delta hedger puts on flatteners
4. Thus, delta hedging stabilizes curve

Figure 9.17: Delta Hedging Profile of a Binary Curve Floor Struck at 0

the low realized vol curve dynamics in 2006.⁴⁰ In aggregate, the delta hedging activities of the many dealers who were long these out-of-the-money binary put options were a significant force. As the curve would steepen, they would come in en masse and put on flatteners, which stabilized the curve. And when the curve would flatten, they would come in and put on steepeners, which again stabilized the curve. These stabilizing forces would be at work, provided the curve did not flatten so much that the options went in-the-money.

However, if the curve were to flatten enough such that the binary floors were in-the-money, the delta hedging activities of someone who is long the

⁴⁰This analysis implicitly assumes that only dealers delta hedged curve options, while buy-side participants who traded them did not. This assumption was indeed generally the case. Note that if both buyers and sellers of curve options were to delta hedge, their hedging would cancel one another out and therefore net hedging activities would have no effect on the curve.

option change dramatically and actually become a destabilizing force for the curve. When the binary floor is in-the-money, as the curve starts to flatten, a delta hedger who is long the floor will actually put on flatteners, not steepeners. On the margin, as the curve is flattening and a delta hedger puts on flatteners himself, this will serve to cause the curve to flatten some more. Conversely, provided the curve is flat enough such that the binary floors are in-the-money, a steepening curve will be met by a delta hedger who needs to put on steepeners himself. This, too, will serve to cause the curve to steepen further in the face of steepening.

The delta hedging profile for a binary curve floor is summarized in Figure 9.17. In this particular case we have assumed that the binary floor is struck at zero. When the curve is positively sloped, the delta hedging activities of someone who is long the option tend to stabilize the curve. But when the curve is inverted, the delta hedging activities of someone who is long the option tend to destabilize the curve.

9.3.8 So Why Did 30-Year Swap Spreads Go Negative — and What Does That Have to Do with Curve Options?

Many thought it an impossibility. Some risked huge sums that it would never happen, and even when it did happen, they entered into further positions that expressed the view that it would soon correct itself. It did correct itself briefly, but then it happened again. And it got worse and worse and worse, and it didn't go away for a long time thereafter.

As we discussed in Chapter 1, 30-year swap spreads went negative for the first time on October 23, 2008, shortly following the collapse of Lehman. This marked the first time that a benchmark swap spread in the dollar market had ever gone negative.⁴¹ There were a number of factors that contributed to this phenomenon, not the least of which was the delta hedging needs of dealer desks on the back of the curve options they had created.

As is often the case in financial markets, there may be a variety of factors at work that dictate market moves. Oftentimes the interplay of these factors is complicated, and figuring out which factor or factors will dominate is a nontrivial task. In the case of 30-year swap spreads in fall 2008, it seems as though there were many factors at work that all conspired to tighten spreads significantly.

Perhaps the most important thing to keep in mind is that the market, in the weeks following the collapse of Lehman, was extremely illiquid. In many cases, people were only trading if they needed to, and even a trade of modest

⁴¹Swap spreads had previously gone negative in other foreign currency swap markets, for example, in the sterling market.

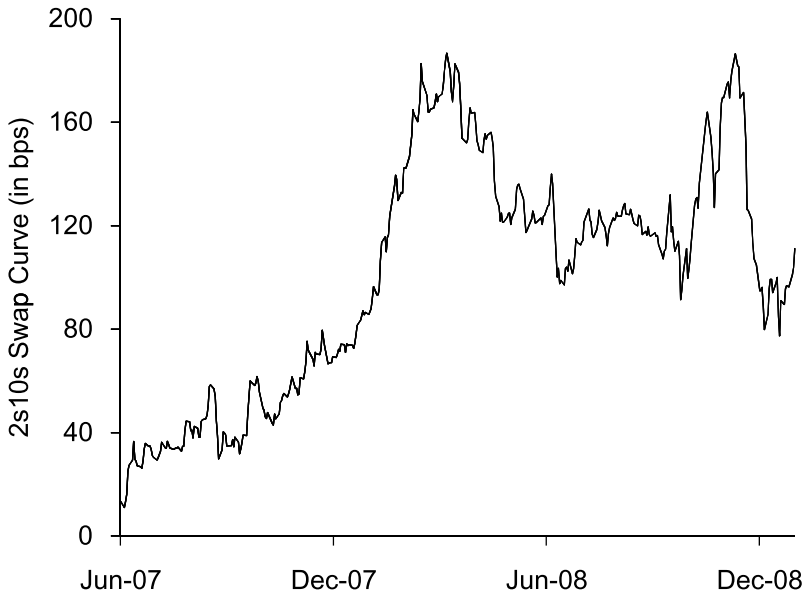


Figure 9.18: The 2s10s Swap Curve Leading Up and Into the Crisis

size could have a sizable impact on the market. As shown in Figure 9.18, as the crisis first started to manifest itself in the summer of 2007, 2s10s started to steepen as the market started to rally, led by the front end of the curve.

When Lehman Brothers fell in 2008, many of its counterparties entertained the possibility of entering into replacement trades with other dealers. This would mean that the counterparties would terminate their existing swap trades with Lehman and go out to the market and replace these trades at levels prevailing in the market at the time. Any losses associated with terminating existing swaps with Lehman would be taken up in bankruptcy courts.⁴²

⁴²Lehman Brothers Holdings Inc. filed for Chapter 11 under the United States Bankruptcy Code on September 15, 2008. A few weeks later, Lehman Brothers Special Financing Inc. (LBSF) filed on October 3, 2008, and two other Lehman entities, Lehman Brothers Financial Products Inc. and Lehman Brothers Derivative Products Inc., filed on October 5, 2008. The initial filing of the holding company in September was enough for many counterparties to examine their ability to terminate their trades with Lehman, regardless of which particular entity that they faced in their swap transactions (the ability of counterparties to terminate their Lehman swap transactions upon the Chapter 11 filing of the holding company was somewhat complicated and depended on which particular Lehman entity a counterparty faced in its swap transactions). Under the standard terms of the ISDA that Lehman had on with many of its counterparties, the filing of Chapter

In particular, pension funds who have long-dated liabilities had received on many long-dated swaps to hedge. When Lehman fell, it was commonly believed that many pension funds went out into the market to put on replacement trades. And as they received fixed in long-dated swaps, primarily in the 30-year sector, this pushed long-dated swap spreads down. In addition, in the midst of the crisis, many financial firms were deleveraging, selling a variety of assets, including long-dated Treasuries. The outright selling of long-dated Treasuries pushed the yields of these treasuries up versus long-dated swaps, which again served to narrow spreads. Some leveraged funds actually had on trades in which they had simultaneously bought long-dated Treasuries and paid fixed on long-dated swaps (i.e., asset swaps). In order to deleverage, these funds sold these Treasuries and essentially received fixed on swaps, which also tended to put downward pressure on long-dated swap spreads. At the same time, as the crisis raged on, many in the market expected that the government would need to radically increase the supply of treasuries to fund the crisis, which also served to narrow swap spreads.⁴³ But yet another major driver of 30-year swap spreads into negative territory had to do with the delta hedging by dealer desks of non-inversion notes on 10s30s that they had underwritten.

Figure 9.19 shows what was going on with 10s30s in swaps as the crisis mounted. Whereas 2s10s steepened significantly in the weeks following the Lehman bankruptcy, 10s30s actually flattened significantly due to the market forces we just discussed. And on October 8, 2008, the 10s30s curve was essentially flat. Now comes the tricky part.

11 constituted an Event of Default, which gave Lehman's counterparties the right, but not the obligation, to terminate their swap transactions. Upon termination, a termination amount based on the mark-to-market of the swaps in question would either be made to or owed by Lehman's counterparties who elected termination. See Stromfeld et al. (2009) for more details.

In one interesting case, a counterparty of LBSF decided not to terminate its swap upon the latter's Chapter 11 filing. The swap was out-of-the-money to the counterparty, and if it had decided to terminate, it presumably would have been required to make a termination payment to LBSF. Not only did the counterparty not terminate the transaction, but (relying on a provision in the ISDA Master Agreement that it claimed released either party from making payments in the event of the other party's default or potential default) it also withheld performance (i.e., it decided not to pay coupons in the swap). However the court ruled that the counterparty could not rely on this provision as a basis for withholding performance. Moreover, the court ruled that the decision by the counterparty not to terminate the swap "promptly upon a debtor counterparty's bankruptcy filing" may be deemed a waiver of this right. This decision suggests that, under the provisions of bankruptcy, a non-defaulting counterparty does not have the right to "play the market" once an Event of Default has occurred. See Davis Polk and Wardwell (2009) for more details.

⁴³In Section 2.2.1, we discussed that an increase or decrease in the supply of Treasuries (or even a change in the expectation of such) may affect swap spreads. An expected increase in the supply of Treasuries tends to lower swap spreads.

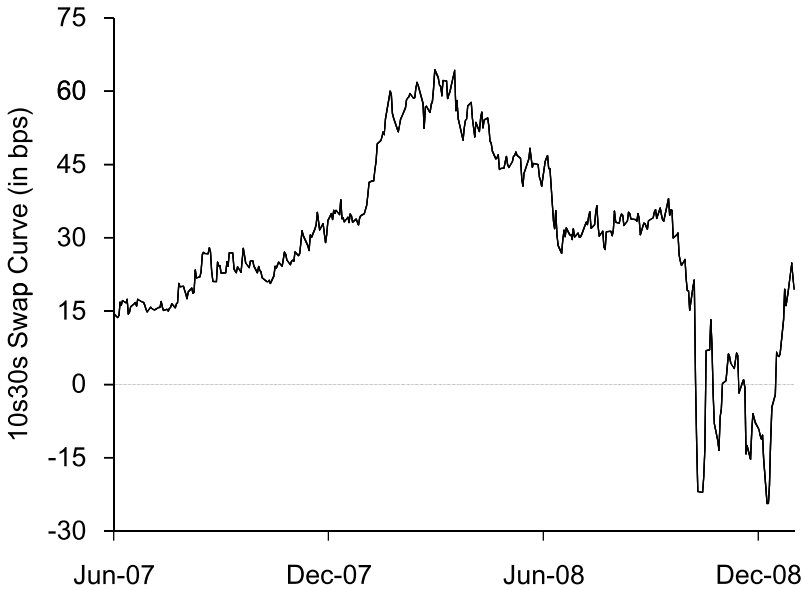


Figure 9.19: The 10s30s Swap Curve Leading Up and Into the Crisis

Dealers had gotten long tons of low-strike binary curve floors on 10s30s through the non-inversion notes that they had underwritten. Many of these were struck at zero. Prior to the crisis when 10s30s was steeper, these options were out-of-the-money, and as the curve started to flatten, these options were approaching being at-the-money. As 10s30s continued to flatten and eventually inverted, many of these options became in-the-money.

As we discussed in Section 9.3.7, when the 10s30s curve inverted and many of the binary curve floors were now in-the-money, exotic desks who were long these options and delta hedging faced a switch in their gamma profile. With the 10s30s curve inverted, these binary curve floors now had negative gamma. That meant that as the curve continued to flatten (i.e., become more inverted), exotic desks were forced to put on 10s30s flatteners. That is, these desks were forced to pay in 10s and receive in 30s. Swap desks across the street were being asked by exotic desks who managed the risk in these 10s30s binary curves to receive in 10s and pay in 30s in order to fulfill these delta hedging requirements. The pace of this became relentless. In particular, as swap desks were asked to receive in 10s and pay in 30s, the 10s30s curve continued to invert, and the 30-year swap spread at which swap dealers were willing to pay went lower and lower and lower, as shown in

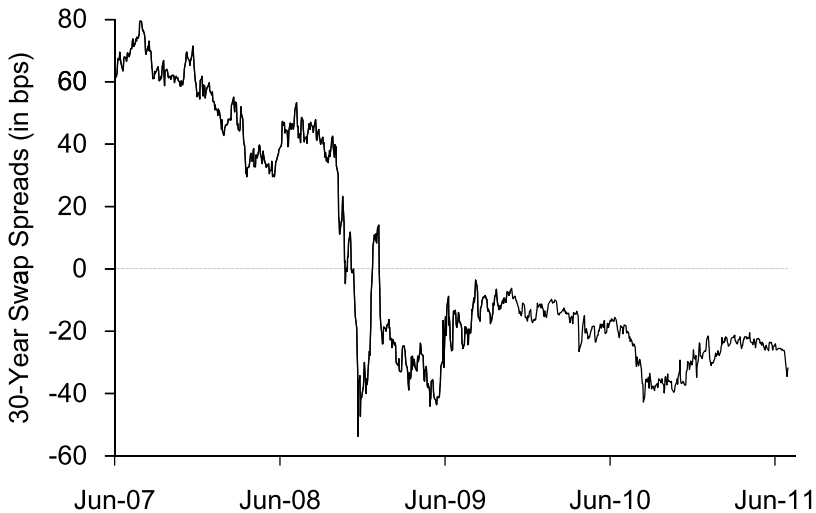


Figure 9.20: 30-Year Swap Spreads Leading Up and Into the Crisis

Figure 9.20. Whereas the 10s30s curve in swaps was essentially flat on October 8, 2008, a few days later on October 13 the curve was 24 bps inverted. Exotic desks who (robotically) needed to delta hedge were undeterred when 30-year spreads were approaching zero; they needed to continue to put on flatteners as the curve continued to flatten, regardless of the levels at which they were getting done. They were like the Terminator: they couldn't be bargained with or reasoned with; they felt no pity or remorse. They would absolutely not stop until they were hedged.

Eventually, 30-year spreads were pushed all the way to zero—and then through zero into negative territory on October 23, 2008. On that date 30-year swap spreads closed at -3 bps. The impossible had happened. Some investors rushed in to buy spreads in negative territory, thinking that this phenomenon would soon correct itself. Some said that they wanted to buy 30-year spreads just so that they could tell their children and grandchildren that they were there when 30-year spreads went negative and that they did not pass up on the opportunity of a lifetime. But those who did probably got burned badly. Thereafter 30-year spreads continued to collapse, and within a few weeks they closed at -53.75 bps. And although 30-year spreads subsequently rebounded into positive territory for a brief time some weeks later (and the 10s30s curve vacillated between being inverted or not), 30-year spreads once again turned negative and remained in negative territory for years thereafter.